#ManyDaughters—Pre-analysis Plan

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Abstract

Using data from the German Socio-Economic Panel (SOEP) from 2000 to 2020, we will study the impact of having daughters on gender attitudes, gender-equal behavior, and political preferences. We will pursue quasi-experimental identification by leveraging the gender of the first-born child to minimize, and where possible, eliminate, the need for strong identifying assumptions and thereby, within our means, facilitate a clean causal interpretation.

Study

As one of many research teams (RTs) participating in a metascience study, we hereby describe our approach for testing a set of predefined hypotheses. We utilize the same data as all other RTs, but are free in applied methodology and in the choice of variables and controls. As of writing this pre-analysis plan, we do not have access to the data. Instead, we will lay out our empirical strategy based on data descriptions from the SOEPcompanion and paneldata.org, and will subsequently gain access to a 5% subsample of the data to develop the analysis code.

In particular, we will study the effect of having daughters on: i) gender norms, represented as attitudes toward gender equality in general and differentiated by sex (men and women), ii) observed behavior regarding gender equality, i.e., self-reported distribution of responsibilities in the household, and iii) political preferences in terms of self-reported placement on a left-right spectrum.

For all hypotheses, we utilize the "gender of the first-born child (daughter/son)" as a quasi-experimental intervention and consider "first-born child (daughter)" as the treatment. In the following, we use "first-born daughter" synonymously with "a daughter born as the first child" to describe said treatment. Where permissible, we will utilize within-household variation, restricting ourselves to those households that were child-free for at least one wave of the panel. We will differentiate between short-, mid-, and long-term effects of having first-born daughters. When the available data are not of longitudinal nature, we will utilize the "gender of first-born child (daughter/son)" as an instrument for "having daughters" to allow for a similar, albeit between-household, identification.

We consider only the gender of the first born child quasi-experimental since the gender of additional children might be the endogenous result of parents' choices. For example, Ichino et al. (2014) find that if the first child is a boy, the likelihood of more children increases. Consistent with the idea that child gender influences fertility decisions, Dahl and Moretti (2008) find that parents with girls are more likely to continue childbearing, suggesting a revealed preference for sons. Similarly, Iacovou (2001) and Lundberg (2005) find that parents with two children of the same sex are more likely to have a third to "balance" the gender composition of their children.

Hypotheses

The following four hypotheses were provided by the organizers of the #ManyDaughters study to be tested by the RTs.

H1: *Having daughters leads to stronger support for gender equality (as measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters)> (X for respondents without daughters)

H2: *Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters)> (Y for respondents without daughters).

H3: *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters)> (Z for respondents without daughters)

H4: *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for male respondents with daughters − X for male respondents without daughters)
> (X for female respondents with daughters − X for female respondents without daughters.)

As a crucial modeling choice, we treat "having daughter**s**" as synonymous with "having **a** daughter", since our identification strategy is based solely on the gender of the first-born child. In contrast to the formulation of the hypotheses, we will not pay special attention to whether a particular household has multiple female children, but will interpret "having daughters" in a conceptual rather than literal sense.[1]

Methods

Data collection

For our analysis, we will use data from the German Socio-Economic Panel (SOEP). The SOEP has been conducting a longitudinal survey study since 1984, currently with about 30,000 active participants Goebel et al. (2019). We will use data from the waves 2000 to 2020 using data included in the SOEP-core data set. Depending on the hypothesis, we will use only a sub-sample of all the participants as detailed below.

Statistical analysis

We follow Benjamin et al. (2018) and use $p < 0.005$ as the threshold for statistically significant evidence while using $p < 0.05$ as suggestive evidence. All our analyses will be based on regression analyses, in particular instrumental variables (IV) regression and multi-period Difference-in-Difference estimation as proposed by Callaway and Sant'Anna (2021).

As a robustness check for our identifying assumption (gender of the first born child), we would ideally exclude couples who either: i) adopted their first child or ii) used in vitro fertilization, as both allow for direct control over the child's gender. However, the SOEP does not contain this information. As robustness checks, we will exclude same-sex couples as a proxy for couples with an increased likelihood of using either non-conventional way of conception (see, e.g, Greenfeld and Seli, 2013).[2]

In the case of longitudinal data, we will exclude households that report an age of the first-born child that exceeds the maximum possible age between the respective waves of data collection, e.g., between waves 2005 and 2009 when the age of the first child is greater than $2009 - 2005 + 1 = 5$. In such a case, the age could have been entered due to a multitude of reasons, including a data entry error, adoption, merging of two households after separation, etc. Most of these reasons would violate our identifying assumption; therefore, we conservatively exclude these observations. Additionally, we will exclude observations for which variables that are used in the analysis are missing due to, for example, item non-response or sample attrition.

[1] On a related note, we appreciate that "many daughters" is a pun that refers to the many RTs contributing to the meta-study.

[2] Moreover, same-sex couples do not get pregnant accidentally, whereby the orthogonality of the gender of the first child and the happenstance of conceiving a child (by accident or by choice) might be violated.

Variables used in statistical analysis

In the following, we describe the variables that we are using to answer each of the four hypotheses.

H1:

As a proxy for gender equality, we will use a three-item construct that has been introduced by Hamjediers (2021) and subsequently used by Stöcker and Schütz (2023). The three items are:
I: *Children under age 6 suffer when their mother works*,
II: *Children under age 3 suffer when their mother works* and,
III: *It is best for the husband and wife to work equally so that they can share family and household responsibilities equally.*

Participants expressed their agreement with these statements on 7-point Likert scales. To construct an index where higher values reflect more gender-equal attitudes, we reverse-code the first two items. We then compute the index by summing all three items with equal weight and normalize the resulting score to range between 0 and 1, with 0 indicating the least and 1 the most gender-equal attitudes. The three corresponding questions were administered only in the 2018 wave. Thus, our analysis will be restricted to participants in that wave who responded to all three items.

We will run an IV regression where the instrument is the gender of the first child (coded as 1 if female zero otherwise) which we assume to be randomly assigned. As stated above, we assume it to be equally likely to conceive a boy or a girl in case of the first child, and we assume parents cannot influence the gender of the child. Thus, the instrument is both exogenous and relevant, as the gender of the first child is strongly correlated with having at least one daughter. This approach extends the empirical strategy in Dahl and Moretti (2008), who treat the gender of the first child as an "as-good-as-random" variable.[3]

The two-stage regressions we will run are specified as:

$$\text{First stage: } \texttt{HasDaughter}_i = \theta_0 + \theta_1 \texttt{FirstChildFemale}_i + \theta_2 \mathbf{Z}_i + u_i$$

$$\text{Second stage: } \texttt{GenderEqualityIndex}_i = \beta_0 + \beta_1 \widehat{\texttt{HasDaughter}}_i + \beta_2 \mathbf{Z}_i + \varepsilon_i,$$

where $\texttt{HasDaughter}_i$ equals 1 if individual i has a daughter and 0 if not. $\texttt{GenderEqualityIndex}_i$ corresponds to the construct as described above. $\widehat{\texttt{HasDaughter}}_i$ are the estimation results from the first stage. $\mathbf{Z}_i$ is a vector of controls that includes: the logarithm of household income, age, gender, education, and employment status of the respondent, the number of children, and a binary indicator for living in East Germany, as these may affect both gender attitudes and the likelihood of having a child (or, in particular, a daughter). Robust standard errors at the individual level will be estimated.

We will consider an estimated effect of β_1 that is statistically significantly different from zero as evidence for **H1**, i.e., that having daughters leads to stronger support for gender equality.

H2:

To answer Hypothesis 2, we will utilize longitudinal data from the SOEP to examine whether individuals' political preferences shift toward the left after having a daughter, compared to their preferences prior to the birth. Specifically, we will compare changes in self-reported political orientation before and after the birth of a first-born daughter, relative to individuals who have not (yet) had a daughter as their first child.

As the dependent variable, we will use the self-reported political orientation, in which participants position themselves on a 11-point Likert scale ranging from left (0) to right (10). This measure is available in the 2005, 2009, 2014, and 2019 waves. For ease of interpretation in line with our hypothesis, we reverse-code the scale so that higher values indicate more left-leaning political preferences. We restrict the sample to individuals who had no children prior to 2005.

We estimate group-time average treatment effects $ATT(g,t)$ using all four survey waves. Each treatment group, defined by the wave in which the first-born daughter was born, is compared to a control group of individuals who had not yet experienced the treatment at that point in time.

[3] Since Dahl and Moretti (2008) analyze the direct effects of the gender of the first child (e.g., on divorce rates and additional childbearing), IV regression is not applicable in their case.

Our treatment variable is a binary indicator that switches from 0 to 1 in the year following the birth of a first-born daughter and remains 1 thereafter. Since the assignment of treatment (i.e., having a daughter as the first child) occurs in different time periods across individuals, and the treatment is absorbing once received, we employ the estimation procedure introduced by Callaway and Sant'Anna (2021), which is specifically designed for staggered treatment adoption in panel settings.

At each point in time, the control group consists of individuals who have not yet had a daughter as their first child, including those who may be treated in future periods (i.e., not-yet-treated). We will report group-time average treatment effects $ATT(g,t)$ based on the time of treatment (i.e., the first wave with a first-born daughter). Given the availability of four waves, this approach allows us to study short-, medium-, and long-term effects of the treatment on political preferences.

Our set of control variables includes: the logarithm of household income, respondent age, gender, education, and employment status (full-time, part-time, or unemployed); the number and gender composition of any additional children; and a binary indicator for residence in East Germany. These variables are included as they may influence both political preferences and the likelihood of having a daughter. Standard errors are clustered at the individual level.

We consider **H2** to be confirmed if the aggregated average treatment effect across groups is positive and significantly different from zero. We consider the hypothesis partially supported if some group-specific estimates are positive and significantly different from zero. In the latter case, the results may suggest that the impact on political preferences forms later (e.g., when children enter primary or secondary school age) or is not persistent.

H3:

To test this hypothesis, we restrict the sample to households in which at least two adults of opposite sexes live. As a proxy for gender-equal behavior in the household, we will use the division of time spent on housework in the years 2000–2020. We calculate the total number of hours the two adults report spending on housework and compute the male-to-female ratio of time spent, serving as our measure of housework division. To assess the effect of treatment, we will exploit the longitudinal nature of the data and examine how this ratio changes when the household is "treated" with a first-born daughter. Specifically, our dependent variable will be the negative absolute deviation from a 50/50 split, such that higher values indicate greater equality.

We compute the group-time average treatment effects $ATT(g,t)$ based on the timing of treatment. Because this outcome is available annually, we leverage the finer temporal granularity of the data to define four time points of treatment-administration:
(1) The initial year the first-born daughter is observed,
(2) Three years after the first-born daughter is observed,
(3) Seven years after the first-born daughter is observed,
(4) Twelve years after the first-born daughter is observed.

Here, (1) is our main specification, while (2) to (4) serve as robustness checks for eventual delayed treatment effects and to mitigate potential confounding factors. Specifically, for (2), we focus on a time when pregnancy cannot factor in, most children attend kindergarten, and maternity/paternity leave has typically ended. Time point (3) corresponds to the typical age at which children enter primary school, while (4) captures the point at which children have transitioned from primary to secondary school across all federal states. The two transitions captured in (3) and (4) are relevant, as these can be moments when parents experience gender bias or discrimination affecting their children.

We will control for: the logarithm of household income, age, gender, education, and employment status (full-time, part-time, or unemployed) of adult household members; the number and gender composition of any additional children; and a binary indicator for residence in East Germany. These variables are included as they may influence both division of housework and the likelihood of having a daughter. Standard errors are clustered at the household level.

We consider **H3** to be confirmed if the aggregated average treatment effect across groups is positive and significantly different from zero in our main specification. We consider the hypothesis partially supported if some estimates of the robustness checks are positive and significantly different from zero, indicating delayed effects.

H4:

To answer **H4** we use the same index for gender equality (again, restricted to the 2018 wave) and the same empirical strategy (IV regression with gender of first-born child as instrument for "having daughter") as in **H1**, with the addition of interaction effects with the respondent's gender. The extended two-stage regressions are specified as follows:

$$\begin{aligned}
\text{First stage: } & \texttt{HasDaughter}_i = \theta_0 + \theta_1 \texttt{FirstChildFemale}_i + \theta_2 \mathbf{Z}_i + u_i \\
\text{Second stage: } & \texttt{GenderEqualityIndex}_i = \beta_0 + \beta_1 \widehat{\texttt{HasDaughter}}_i + \beta_2 \texttt{MaleRespondent}_i \\
& + \beta_3 \left(\widehat{\texttt{HasDaughter}}_i \times \texttt{MaleRespondent}_i \right) + \beta_4 \mathbf{Z}_i + \varepsilon_i,
\end{aligned}$$

where $\texttt{MaleRespondent}_i$ is a binary indicator equal to 1 if the respondent is male and 0 if female. The interaction term $\widehat{\texttt{HasDaughter}}_i \times \texttt{MaleRespondent}_i$ captures heterogeneous treatment effects by the respondent's gender. As before, $\mathbf{Z}_i$ includes the same set of control variables, with the exception of gender, which is modeled separately and interacted with the treatment indicator. Robust standard errors at the individual level will be estimated.

We consider **H4** to be confirmed if the interaction effect β_3 in the second stage regression is positive and significantly different from zero.

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